

MODEL QUESTION PAPER1
S3 INSTRUMENTATION ENGINEERING
SUBJECT: DIGITAL CIRCUITS AND SYSTEM

Time: 3 Hours

Max. Marks: 75

I: Part – A: Answer *all* the following questions in one word or sentence. **(9 x 1 = 9 Marks)**

1. List the different types of number system.
2. Define base of a number system
3. Draw the truth table of Half adder.
4. Define multiplexer
5. List any two applications of flip flops..
6. List any two types of shift register
7. Define resolution of DAC
8. List any two types D/A converter
9. Define settling time of DAC

II: Part - B: Answer *any eight* questions. Each question carries 3 marks. **(8 x 3 = 24 Marks)**

1. Realize AND , OR and EX OR using NAND gates
2. State Demorgan's theorem
3. Describe diagram of 4X1 multiplexer .
4. Draw the diagram of 3 bit binary to grey converter
5. Describe the working of SR flip flop
6. Explain the working of 3 bit ring counter
7. Compare synchronous and asynchronous counter
8. Draw the diagram of R-2R ladder type DA converter
9. List the advantages of K map
10. Describe the working of SISO shift register

Part - C: Answer *all* questions. Each question carries 7 marks. **(6 x 7 = 42 Marks)**

III. Implement the logic functions using basic gates

$$Y = ABC + ABC + AB C + ACD$$

OR

IV. Using Boolean Algebra simplify $ABC [AB + C(BC + AC)]$

V. Design a Full ADDER using EX-OR and Basic Gates

OR

VI. Draw the logic diagram for 2 to 4 Decoder and explain its operation

VII. Draw and explain the logic diagram of a clocked JK flip flop

OR

VIII. Explain working of 4 bit Johnson counter.

IX. Describe the working of binary weighted resistor DAC

OR

X. Describe the working of successive approximation type ADC

XI. Design synchronous mod-6 counter .

OR

XII. Design asynchronous mod-10 counter

XIII. With necessary diagram explain 1 to 4 de multiplexer

OR

XIV. Explain BCD to decimal decoder

Prepared by:
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Scrutinized by:

Chairman:
Designation:
Institution:

Question Wise Analysis

Course : **DIGITAL CIRCUITS AND SYSTEMS – QP 1**

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	REMEMBERING	1	2
I.2	M1.01	REMEMBERING	1	2
I.3	M2.01	REMEMBERING	1	2
I.4	M2.03	REMEMBERING	1	2
I.5	M3.05	REMEMBERING	1	2
I.6	M3.04	REMEMBERING	1	2
I.7	M4.04	REMEMBERING	1	2
I.8	M4.01	REMEMBERING	1	2
I.9	M4.04	REMEMBERING	1	2
II.1	M1.02	APPLYING	3	9
II.2	M1.03	REMEMBERING	3	9
II.3	M2.03	UNDERSTANDING	3	9
II.4	M2.06	UNDERSTANDING	3	9
II.5	M3.02	UNDERSTANDING	3	9
II.6	M3.05	UNDERSTANDING	3	9
II.7	M3.06	UNDERSTANDING	3	9
II.8	M4.02	UNDERSTANDING	3	9
II.9	M1.03	REMEMBERING	3	9
II.10	M3.04	UNDERSTANDING	3	15
III	M1.02	APPLYING	7	15
IV	M1.03	APPLYING	7	15
V	M2.01	UNDERSTANDING	7	15
VI	M2.04	UNDERSTANDING	7	15
VII	M3.02	UNDERSTANDING	7	15
VIII	M3.05	UNDERSTANDING	7	15
IX	M4.01	UNDERSTANDING	7	15
X	M4.03	UNDERSTANDING	7	15
XI	M3.07	APPLYING	7	15

XII	M3.06	APPLYING	7	15
XIII	M2.03	UNDERSTANDING	7	15
XIV	M2.04	UNDERSTANDING	7	15

Course : **DIGITAL CIRCUITS AND SYSTEMS– QP 1**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	11	30	2	2	3	9	2	14	7	25
2	14	38	2	2	2	6	4	28	8	36
3	14	38	2	2	4	12	4	28	10	42
4	6	17	3	3	1	3	2	14	6	20
Total	45	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	15	12.1
Understanding	77	62.6
Applying	31	25.2
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

MODEL QUESTION PAPER2
S3 INSTRUMENTATION ENGINEERING
SUBJECT: DIGITAL CIRCUITS AND SYSTEM

Time: 3 Hours

Max. Marks: 75

I: Part – A: Answer all the following questions in one word or sentence. (9 x 1 = 9 Marks)

1. Give two examples of un weighted code.
2. Define positive logic
3. If A, B, C are inputs, what will be sum and carry terms of a full adder
4. List the applications of multiplexer
5. List any two applications of shift register
6. List the applications of counter
7. Define monotonicity of DAC
8. List any two ADC
9. Define modulus of a counter

II: Part - B: Answer any eight questions. Each question carries 3 marks. (8 x 3 = 24 Marks)

1. Realize AND, OR and EX OR using NOR gates
2. Convert 127_8 and $B86_{16}$ into decimal
3. Describe diagram and truth table of 1X4 de multiplexer
4. Draw the diagram and truth table of 3 bit to grey to binary converter
5. Describe the working of D flip flop
6. Draw the diagram and counting sequence of 3 bit Johnson counter
7. Draw the diagram and truth table of BCD to decimal decoder
8. Draw the diagram of binary weighted DA converter
9. List the disadvantages of K map
10. Describe the working of PIPO shift register

Part - C: Answer all questions. Each question carries 7 marks. (6 x 7 = 42 Marks)

- III. Convert the following decimal numbers to hexadecimal numbers
i) $(718.25)_{10}$ ii) $(385.5)_{10}$
OR
- IV. Using K map simplify $F(A,B,C,D) = \sum m(6,7,8,10,13,15) + \sum d(0,1,2,3)$
- V. Design a Full subtractor using EX-OR and Basic Gates
OR
- VI. Design 4 bit gray to binary code converter

VII. Draw and explain the operation of a master slave JK flip flop

OR

VIII. Explain working of 4 bit ring counter.

IX. Describe the working of R-2R ladder type DAC

OR

X. Describe the working of counter type ADC

XI. Design synchronous mod- 8 counter .

OR

XII. Explain asynchronous 3bit up down counter

XIII. With necessary diagram explain 4 to1 multiplexer

OR

XIV. Explain 4 bit binary to gray code converter

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Question Wise Analysis

Course: **DIGITAL CIRCUITS AND SYSTEM – QP 2**

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	REMEMBERING	1	2
I.2	M1.01	REMEMBERING	1	2
I.3	M2.01	UNDERSTANDING	1	2
I.4	M2.03	REMEMBERING	1	2
I.5	M3.04	REMEMBERING	1	2
I.6	M3.06	REMEMBERING	1	2
I.7	M4.07	REMEMBERING	1	2
I.8	M4.03	REMEMBERING	1	2
I.9	M3.06	REMEMBERING	1	2
II.1	M1.02	APPLYING	3	9
II.2	M1.01	APPLYING	3	9
II.3	M2.03	UNDERSTANDING	3	9
II.4	M2.05	UNDERSTANDING	3	9
II.5	M3.02	UNDERSTANDING	3	9
II.6	M3.05	UNDERSTANDING	3	9
II.7	M2.04	UNDERSTANDING	3	9
II.8	M4.01	UNDERSTANDING	3	9
II.9	M1.03	REMEMBERING	3	9
II.10	M3.04	UNDERSTANDING	3	9
III	M1.01	APPLYING	7	15
IV	M1.03	APPLYING	7	15
V	M2.02	UNDERSTANDING	7	15
VI	M2.05	APPLYING	7	15
VII	M3.03	UNDERSTANDING	7	15
VIII	M3.05	UNDERSTANDING	7	15
IX	M4.02	UNDERSTANDING	7	15
X	M4.03	UNDERSTANDING	7	15
XI	M3.07	APPLYING	7	15
XII	M3.08	UNDERSTANDING	7	15
XIII	M2.03	UNDERSTANDING	7	15
XIV	M2.06	UNDERSTANDING	7	15

Course: **DIGITAL CIRCUITS AND SYSTEM – QP 2**

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Mark Distribution

Module	Hr/Module	Marks\Module (hi\ΣHi)*123(±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	11	30	2	2	3	9	2	14	7	25
2	14	38	2	2	3	9	4	28	9	39
3	14	38	3	3	3	9	4	28	10	40
4	6	17	2	2	1	3	2	14	5	19
Total	45	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	11	9
Understanding	78	64
Applying	34	27
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100